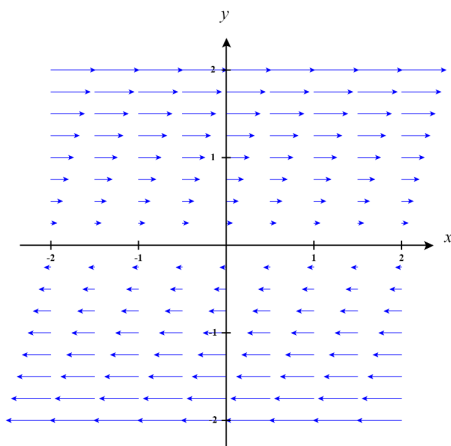


Math 2551 Worksheet: Curl, Divergence, Green's Theorem

1. Below is a plot of a vector field $\mathbf{F}(x, y)$. Use this to decide whether the values of $\text{curl } \mathbf{F} \cdot \mathbf{k}$ and $\text{div } \mathbf{F}$ in each quadrant are positive, negative, or zero.



2. Compute $\text{div } \mathbf{F}$ and $\text{curl } \mathbf{F} \cdot \mathbf{k}$ for the vector field $\mathbf{F}(x, y) = \langle \frac{y}{4}, 0 \rangle$, which was plotted above.

3. Let C be the ellipse

$$\left(\frac{x}{3}\right)^2 + \left(\frac{y}{4}\right)^2 = 1.$$

- (a) Parametrize this ellipse to give it a positive orientation.
- (b) Let $\mathbf{F}(x, y) = 2x\mathbf{i} + 2y\mathbf{j}$. Use Green's theorem to find the circulation of $\mathbf{F}$ around C and its flux across C .
4. Let R be the region in the xy -plane bounded above by the curve $y = 3 - x^2$ and below by the curve $y = x^4 + 1$. Orient this boundary positively. Let

$$\mathbf{F}(x, y) = (y + e^x \ln y)\mathbf{i} + (e^x/y)\mathbf{j}.$$

Use Green's theorem to find the circulation of $\mathbf{F}$ around C . What happens when you try to use Green's theorem to evaluate the flux of $\mathbf{F}$ across C ? Should you use Green's theorem to evaluate the flux integral?

5. Use Green's Theorem to find the work done by the force $\mathbf{F}(x, y) = \langle x(x + y), xy^2 \rangle$ in moving a particle from the origin along the x -axis to $(1, 0)$, then along the line segment to $(0, 1)$, and then back to the origin along the y -axis.
6. **Looking ahead:** Find a parameterization (a function $\mathbf{r}(s, t) = \langle x(s, t), y(s, t), z(s, t) \rangle$) of the plane through the origin that contains the vectors $\mathbf{i} - \mathbf{j}$ and $\mathbf{j} - \mathbf{k}$. Linear algebra ideas may be useful.